Finite-width bounds for the globally margined tanh posterior

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Abstract

We give finite-width bounds for the actual Gaussian tanh posterior conditioned on absolute margin greater than 1/2 at every Boolean input and exact training labels. A bounded-weight, globally margined width-C teacher at depth L is learned at every width $N \geq C$ with sample size essentially $[Nd + LN^2]L \log(d + C + L + B + N)$, uniformly over canonical variance prefactors $0 < t \leq 1$. The proof combines a teacher neighborhood, a conditional energy tail, and a margin-stable parameter net. For bounded-fan-in Boolean circuit targets, a separate Gaussian mean-shift witness improves this to a linear-width bound $O(N(d + CH)/t^2)$ above an explicit width threshold, where $H = L - 1$ and C counts the circuit wires. This second bound retains its inverse-scale dependence. Conversely, augmented-coordinate symmetry forces $\Omega(\log d)$ samples for a constant target, at every finite width and depth. We also exhibit a two-dimensional example in which correct-label posterior probability strictly decreases when width increases from one to two. These are finite-width statements. They neither remove the learner-width dependence from the upper bound nor establish an exponential-in-depth lower bound.

1 Model and confidence criterion

Let $X_d = \{-1, 1\}^d$, $d \geq 1$, with augmented input $(1, x)$. There are $L \geq 2$ matrices: $L-1$ tanh hidden layers, each of width N, and a linear readout $z_W(x)$. There are no internal biases. Independent Gaussian entries have variances

$$t^2/d, \qquad t^2/N, \qquad t^2/N^2$$

in the first matrix, internal matrices, and readout, respectively. The canonical range is $0 < t \leq 1$. Write μ for this raw Gaussian law. For a labeled sample $S = ((x_i, f(x_i)))_{i=1}^n$, the posterior is

$$Q_S = \mu(\,\cdot \mid E_S), \qquad E_S = \{|z_W(x)| > 1/2 \ \forall x \in X_d\} \cap \{\operatorname{sign} z_W(x_i) = f(x_i) \ \forall i\}.$$

This is exactly the result of conditioning first on global absolute margin and then on the labels. Repeated examples add no new constraints.

For an input law D, put

$$p_S(x) = Q_S\{\operatorname{sign} z_W(x) = f(x)\}, \qquad e(S) = \mathbb{P}_{x \sim D}\{p_S(x) \leq 2/3\}.$$

The upper bound below controls $e(S)$ with high probability over an iid sample. Its teacher hypothesis is a width-C representation at the same depth L, with physical weights bounded by $B \geq 1$ and signed output at least one everywhere. A margin-1/2 teacher is handled by doubling its readout and replacing B by $2B$.

2 A sample upper bound quadratic in learner width

Theorem 2.1. *There is a universal c such that, for every teacher as above, every $N \geq C$, every $0 < t \leq 1$, every input law D, and $0 < \epsilon, \delta < 1$, one has $\mathbb{P}_S\{e(S) \leq \epsilon\} \geq 1 - \delta$ whenever*

$$n \geq \frac{c}{\epsilon}\left[pL\log(d + C + L + B + N + \epsilon^{-1} + 2) + \log(1/\delta)\right], \qquad p = N(d+2) + (L-2)N^2.$$

The estimate is uniform in prefactor schedules and in limiting strong-prior posteriors at fixed architecture.

2.1 Energy coordinates and a stable teacher neighborhood

Use coordinates

$$A_1 = \sqrt{d/N}\, W_1, \qquad A_j = W_j \ (2 \leq j < L), \qquad A_L = \sqrt{N}\, v.$$

Their p entries are independent $N(0, \sigma^2)$ with $\sigma^2 = t^2/N \leq 1/N$. Write $\kappa(A) = \sum_j \|A_j\|_F^2 = \|A\|_2^2$. Equivalently, the network takes input $u_0 = (1, x)/\sqrt{d}$ and applies each hidden matrix followed by

$$\phi_N(s) = \tanh(\sqrt{N}s)/\sqrt{N}.$$

This activation fixes zero and is 1-Lipschitz, and $\|u_0\| \leq \sqrt{2}$. The last layer remains linear.

Lemma 2.2. *The teacher has an exactly equivalent width-N realization A_* with*

$$\kappa(A_*) \leq K_0 := 2B^2\{d(d+1) + (L-1)C^2\}.$$

On the energy ball $\kappa(A) \leq R$, the output at every input is Lipschitz in the Euclidean parameter norm, with constant

$$\Lambda_R = \sqrt{2L}\max\{1, R/(L-1)\}^{(L-1)/2}.$$

Proof. Pad the teacher to width C, and replicate each hidden neuron either $\lfloor N/C \rfloor$ or $\lceil N/C \rceil$ times. Retain its incoming weights and divide outgoing weights by its replication count. Outputs are preserved exactly. The weighted first and last energies increase by at most two; each internal squared Frobenius norm increases by at most the largest ratio of two replication counts, also at most two. At width C, the three energy contributions are at most $d(d+1)B^2$, $(L-2)C^2B^2$, and C^2B^2, proving the first claim.

The derivative with respect to matrix block j has Frobenius norm at most $\sqrt{2}$ times the product of the other $L-1$ block norms. Arithmetic–geometric mean bounds that product by $(R/(L-1))^{(L-1)/2}$. Sum the squared block derivative bounds and integrate along the segment joining two parameters. The energy ball is convex, so the same bound holds on the entire segment. □

Set $K = 2K_0 + 2$ and $r = 1/(4\Lambda_K) \leq 1$. The Euclidean r-ball around A_* lies in the energy-K ball, and every parameter in it has signed output at least $3/4$ at every input. Thus it belongs to E_S for every sample labeled by f.

2.2 A conditional energy tail at every width

Inside this teacher neighborhood take the coordinate cube with half-side $h = \min\{r/\sqrt{p}, \sigma\}$. Its Gaussian density is at least $(2\pi\sigma^2)^{-p/2}e^{-K/(2\sigma^2)}$, so

$$\mu(E_S) \geq e^{-K/(2\sigma^2)-pT}, \qquad T = \log\left[\sqrt{\pi/2}\max\{1, \sqrt{p/N}/r\}\right]. \tag{1}$$

Indeed its volume is $(2h)^p$, and $\sigma\sqrt{p} \leq \sqrt{p/N}$. Clipping the cube at σ makes T independent of the prefactor; there is no $\log(1/t)$ loss.

The elementary Gaussian Chernoff bound gives

$$\mu\{\kappa(A) > R\} \leq 2^{p/2}e^{-R/(4\sigma^2)}.$$

For $0 < \beta < 1$, define

$$R_\beta = 2K + \frac{4}{N}\left[p\{T + \tfrac{1}{2}\log 2\} + \log(1/\beta)\right]. \tag{2}$$

Divide the tail bound by (1), and use $\sigma^2 \leq 1/N$, to obtain the sample-uniform posterior estimate

$$Q_S\{\kappa(A) > R_\beta\} \leq \beta. \tag{3}$$

Both conditioning events remain present in this calculation.

Writing $s = d + C + L + B + 2$, the definitions imply

$$T \leq c[L\log s + \log(d + LN + 2)],$$

$$R_\beta \leq c\left[K + (d + LN)\{L\log s + \log(d + LN + 2)\} + N^{-1}\log(1/\beta)\right].$$

In particular this radius is polynomial in the ordinary resources and N, uniformly over all canonical scales.

2.3 Margin-stable signs and posterior confidence

Cover the energy-R_β ball by a Euclidean net of radius $1/(8\Lambda_{R_\beta})$, with centers in the ball. The number of centers satisfies

$$M \leq (1 + 16\sqrt{R_\beta}\Lambda_{R_\beta})^p, \qquad \log M \leq cpL\log(s + N + \beta^{-1}).$$

Every globally clean-margin network in this ball has the same signs on the entire cube as a net center: the uniform output change is at most $1/8$. These centers therefore define a fixed class of at most M truth tables, before the sample or input law is chosen. Centers with a zero output may use any fixed sign convention; those selected for a clean-margin network have no such zero.

A finite-class union bound shows that, with sample probability at least $1 - \delta$, every center fitting the labels has population error at most α, provided $n \geq [\log M + \log(1/\delta)]/\alpha$. On this event all low-energy posterior networks have error at most α. Equation (3) bounds the remaining posterior mass by β, so posterior Gibbs error is at most $\alpha + \beta$. Every point counted by $e(S)$ contributes at least $1/3$ to Gibbs error. Thus $e(S) \leq 3(\alpha + \beta)$. Taking $\alpha = \beta = \epsilon/6$ proves Theorem 2.1.

The sample event and the Gibbs bound hold uniformly in t. On the finite truth-table simplex, Gibbs error passes continuously to any limiting strong-prior posterior. Applying the same final inequality proves the stated limiting version without asserting continuity of the confidence-threshold event itself.

Corollary 2.3. *If d, C, B are polynomially bounded in L, and a fixed positive accuracy/confidence requirement needs at least e^{cL} samples along a canonical finite-width sequence, then*

$$\log N \geq cL/2 - O(\log L).$$

Proof. If $\log N > cL$ there is nothing to prove. Otherwise the sufficient sample bound in Theorem 2.1 is N^2 times a polynomial in L. Rearranging gives the claim. □

3 A linear-width bound for Boolean circuit targets

For targets specified by digital gates, a direct Gaussian mean shift improves the preceding width dependence. This theorem retains an explicit t^{-2} factor; it is not a uniform cold-limit bound. Write $H = L - 1$. Suppose f has a layered AND/OR/NOT/COPY circuit with at most H stages and width at most C, including a constant $+1$ wire. Pad shorter circuits by COPY gates.

Theorem 3.1. *If*

$$N \geq 1600C\{d \log 2 + \log(8CH)\},$$

then the raw Gaussian prior satisfies

$$\mu\{f(x)z_W(x) > 1/2 \ \forall x \in X_d\} \geq \exp[-2002N(d + CH)/t^2].$$

For any common training and test input law, the actual posterior therefore has confidence-failure mass at most ϵ with sample probability at least $1 - \delta$ whenever

$$n \geq \frac{6}{\epsilon}\left[\frac{2002N(d + CH)}{t^2} + \log\frac{6}{\epsilon\delta}\right].$$

Proof. Partition each hidden layer into C groups whose sizes n_g are $\lfloor N/C \rfloor$ or $\lceil N/C \rceil$, hence $n_g \geq N/(2C)$. A group represents one circuit wire by its average activation. Use gain $a = 10$ and the affine gate formulas

$$u + v - 1, \quad u + v + 1, \quad -u, \quad u$$

for AND, OR, NOT, and COPY. A constant term is supplied by the actual previous constant group, or by the augmented input at the first layer. Coincident inputs may be combined; the coefficient absolute sum remains at most three.

Define a product Gaussian comparison law Q with the original variances and deterministic shifted means. A first-layer gate row has mean a times its affine formula's coefficients. At later layers, put mean ac_j/n_j on every coordinate of predecessor group j, where c_j is that formula's coefficient. All other weight means are zero. The constant group copies its predecessor constant, starting from the augmented constant input. Set the readout mean to $1/n_{\text{out}}$ on the output group and zero elsewhere. All background random weights remain present.

Suppose the predecessor group averages have the correct signs and signed magnitudes at least 0.9. Since all tanh values lie in $[-1, 1]$, AND and OR retain signed affine margin at least $2(0.9) - 1 = 0.8$, including error in the constant group. COPY and NOT retain margin at least 0.9. The first-layer margin is at least one. Thus each controlled row has signed preactivation mean at least eight. Conditional on previous layers its variance is at most two initially and at most one later, uniformly for $t \leq 1$. For centered Gaussian Z of variance at most two,

$$\mathbb{E}\tanh(8 + Z) \geq 1 - 2e^{-4} - 2e^{-8} > 0.95.$$

Indeed $\mathbb{P}(Z < -4) \le e^{-4}$ and $\tanh(8+Z) \ge 1 - 2e^{-8}$ on the complementary event. Symmetry treats a negative gate bit.

Conditional Hoeffding concentration gives probability at most $e^{-n_g/800}$ that one signed group average is below 0.9 at one input. Sequential union bounds over all layers, groups, and cube inputs give hidden propagation failure at most

$$CH2^d e^{-N/(1600C)} \le 1/8.$$

No independence between distinct cube inputs is used. Conditional on the hidden layers, centered readout noise has variance at most t^2/N at each input, so

$$Q\{\max_x |z(x) - \mathbb{E}[z(x) \mid \text{hidden layers}]| > 1/4\} \le 2^{d+1} e^{-N/(32t^2)} \le 1/8.$$

On the joint good event every signed output is at least $0.9 - 0.25 = 0.65$. Thus $Q(E_f) \ge 3/4$ for the globally correct margin event E_f in the theorem.

We charge every mean shift. The first-layer relative entropy is at most $450dN/t^2$. An internal row has squared mean norm at most

$$a^2 \sum_j c_j^2/n_j \le 18a^2 C/N,$$

so its entropy is at most $900C/t^2$. The readout contributes at most NC/t^2. Hence

$$\mathrm{KL}(Q\|\mu) \le \frac{N}{t^2}\{450d + 900C(H-1) + C\} \le \frac{1000N(d+CH)}{t^2}.$$

For any event E, binary relative entropy gives

$$\mathrm{KL}(Q\|\mu) \ge Q(E)\log(1/\mu(E)) - \log 2.$$

Using $Q(E_f) \ge 1/2$ proves the raw target-mass bound with constant 2002.

Let π be the prior already conditioned on global absolute margin. Since E_f is contained in that event, its truth-table mass under π is at least $p_* := \exp[-2002N(d+CH)/t^2]$. This is a lower bound for every target-labeled sample's version-space normalizer. The expected π-mass of classifiers fitting an iid sample and having population error greater than $\epsilon/6$ is at most $e^{-n\epsilon/6}$. Markov therefore gives

$$\mathbb{P}_S\{Q_S(\mathrm{err} > \epsilon/6) > \epsilon/6\} \le \frac{6e^{-n\epsilon/6}}{\epsilon p_*} \le \delta.$$

Outside this exceptional sample event the posterior Gibbs error is at most $\epsilon/3$. A point with correct-label probability at most $2/3$ contributes at least $1/3$ to that Gibbs error, so its total probability mass is at most ϵ. This argument uses the actual posterior mixture and both conditioning events. □

Remark 3.2 (Layerwise resources and exponent scaling). If the circuit widths are $C_1, \dots, C_H$, put $C_{\max} = \max_j C_j$ and $G = \sum_j C_j$, including all constant and padding COPY wires. Separate balanced partitions at each layer give internal entropy at most $900NC_{j-1}/t^2$ at stage j, and readout entropy at most NC_H/t^2. Thus replace CH by G in the theorem and its width threshold by

$$1600C_{\max}\{d\log 2 + \log(8G)\}.$$

Under the common exponent convention $t = N^{(1-\gamma)/2}$, the width factor N/t^2 equals N^γ. One may take the smaller of this bound and Theorem 2.1. At $\gamma = 1$ the new dependence is linear in N above the displayed threshold. For $C_{\max} \le N$ below that threshold, Theorem 2.1 and the bounded-weight digital realization give a coarser N times polynomial-circuit-resource bound, since the extra factor N is then polynomially bounded. This comparison does not assert the sharper coefficient $d + G$ below the threshold.

The polynomial-depth statement here concerns a digital Boolean implementation at the available learner depth. A tanh teacher with soft intermediate states need not have such an implementation at comparable resources. Nor does the explicit t^{-2} factor permit taking a cold limit in this particular bound. Exponentially large learner widths can still make its sample bound exponential in depth.

4 An exact symmetry lower bound for a constant target

Here $f \equiv +1$ and D is uniform on the cube. The argument works for every positive prefactor t, every finite width and depth, and more generally whenever the first-layer law is invariant under signed permutations of its augmented input columns and is independent of the remaining layers. Only odd hidden activations and a linear readout without internal biases are required, with a conditioning event of positive probability.

For a confidence threshold $c > 1/2$, write $e_c(S) = \mathbb{P}_x\{Q_S(\operatorname{sign} z_W(x) = +1) \le c\}$. A sample coordinate column is *constant* if all its observed values are the same sign.

Theorem 4.1. *If a nonempty sample has k constant coordinate columns, then*

$$e_c(S) \ge \frac{1}{2} - 2^{-k-1}\binom{k}{\lfloor k/2 \rfloor} \qquad (c > 1/2).$$

For $n \ge 1$ uniform iid examples, their number satisfies $K \sim \mathrm{Binomial}(d, 2^{1-n})$. In particular

$$\mathbb{P}_S\{e_c(S) \ge 1/4\} \ge 1 - (1 - 2^{1-n})^d.$$

For fixed $0 < \epsilon < 1/2$ and $0 < \delta < 1$, obtaining $e_c(S) \le \epsilon$ with probability at least $1 - \delta$ requires $n \ge \log_2 d - O_{\epsilon,\delta}(1)$.

Proof. First suppose coordinate j has observed value $s \in \{-1, 1\}$. Let A exchange augmented coordinates $0, j$ and multiply both by s. Define a cube involution T by

$$(Tx)_j = x_j, \qquad (Tx)_\ell = s x_j x_\ell \quad (\ell \ne j).$$

Then $A(1, x) = s x_j (1, Tx)$. The whole network is odd in its augmented input, so applying A to the first-layer columns transforms its output to $z_A(x) = s x_j z(Tx)$. This preserves the raw prior and the global absolute-margin event. At every observed point $x_j = s$, it also has $Tx = x$ and $s x_j = 1$, preserving every training constraint. Thus it preserves the exact posterior, including all its parameter components. If $p(x) = Q_S\{z_W(x) > 0\}$, then on $x_j = -s$,

$$p(Tx) = 1 - p(x).$$

The involution preserves uniform measure on that half cube. Since $c > 1/2$, at most half of it can have $p > c$, giving $e_c(S) \ge 1/4$.

For the full bound, flip the k constant columns so all observed values become $+1$. This preserves the uniform input law and the constant target. Permutations of these columns and augmented column 0 preserve every training constraint. Permuting the k ordinary columns shows that the posterior vote depends on them only through their number r of negative signs, and on the outside coordinates y. Call it $p_r(y)$, and let $w_r = 2^{-k}\binom{k}{r}$. For $1 \le r \le k$, exchange column 0 with one negative coordinate and use oddness to renormalize the augmented first coordinate. This gives

$$p_{k+1-r}(-y) = 1 - p_r(y). \tag{4}$$

For two distinct paired levels, their total failure mass is at least $\min(w_r, w_{k+1-r}) = \min(w_r, w_{r-1})$. At a self-paired middle level, (4) gives failure on at least half the level; if no outside coordinates remain, it forces $p_r = 1/2$ and only strengthens this estimate. Counting every pair twice yields

$$e_c(S) \geq \frac{1}{2}\sum_{r=1}^{k} \min(w_r, w_{r-1}) = \frac{1}{2}(1 - \max_r w_r).$$

The last identity follows from unimodality of the binomial weights. For $k = 0$ the stated bound is zero.

Under uniform iid sampling the d coordinate columns are independent, and each is constant with probability 2^{1-n}. This proves the binomial and single-column assertions. For the final statement choose a fixed k_0 making the displayed error bound greater than ϵ. If $d2^{1-n}$ exceeds a sufficiently large constant depending on k_0, δ, a binomial Chernoff bound makes $K \geq k_0$ with probability greater than δ. Rearranging proves the necessity. □

A stronger asymptotic form follows directly. If $n \leq (1-\eta)\log_2 d$ for fixed $\eta > 0$, then with probability at least $1 - e^{-d^\eta/4}$ one has $K \geq d^\eta$, and hence

$$e_c(S) \geq 1/2 - O(d^{-\eta/2}).$$

The constant target has a bounded-weight realization of width at most two at every depth: use the augmented constant input, carry a positive scalar with weight two through each hidden layer, and use readout weight two. Its activations stay above $1/2$ and its output exceeds one. Thus the lower bound already applies to a simple globally margined teacher.

Remark 4.2 (Scope of the symmetry argument). This concerns correct-label confidence, not an unspecified tie convention for ordinary majority prediction. Its lower bound is logarithmic in dimension. The group of signed augmented-coordinate permutations has at most $2^{d+1}(d+1)!$ elements. For any target, its orbit under these input-label actions is therefore a concept class of that size, and ordinary realizable consistency controls this orbit after $O((d\log(d+2) + \log(1/\delta))/\epsilon)$ samples. This does not bound the entire neural posterior. It explains why indistinguishability within this exact symmetry orbit alone cannot yield exponential-depth hardness when d is polynomial in depth.

5 A caution against pointwise monotonicity in width

Proposition 5.1. *For the actual Gaussian posterior at positive finite scales, correct-label posterior probability at an unseen input need not be nondecreasing with learner width. No finite-factor domination of larger-width wrong-label probability by smaller-width wrong-label probability holds uniformly.*

Proof. Take $d = 2$, $L = 2$, and target $f(x) = \mathrm{sign}(1 + x_1 + x_2)$. The width-one teacher $z_f(x) = \tanh(1 + x_1 + x_2)$ has all weights one and strict global margin greater than $1/2$. Observe the three labels

$$(1,-1) \mapsto +1, \qquad (-1,1) \mapsto +1, \qquad (-1,-1) \mapsto -1.$$

The only completions are f and $g(x) = -x_1x_2$. A width-one one-hidden-layer tanh network has an affine-threshold sign function, whereas g is not an affine threshold. Consequently the width-one posterior assigns probability one to the correct label at $(1,1)$. Its conditioning event has positive probability by a strict-margin neighborhood of the displayed teacher.

At width two, the network

$$z_g(x) = 3\tanh(10(1 + x_1 + x_2)) - 6\tanh(0.4 + 0.2x_1 + 0.2x_2)$$

represents g with strict global margin greater than $1/2$. For $s = x_1 + x_2 = -2, 0, 2$, its values are respectively

$$-3\tanh 10, \qquad 3\tanh 10 - 6\tanh 0.4, \qquad 3\tanh 30 - 6\tanh 0.8.$$

The first is below -2.97, the second exceeds 0.57, and the third is below $-97/125$. These inequalities follow from $\tanh 10 > 0.99$, $\tanh u \le u$, and $\tanh u \ge u - u^3/3$ for $u \ge 0$. The function f also has a strict-margin width-two realization by zero padding. Both therefore have open parameter neighborhoods of positive Gaussian mass at every finite positive scale. After global margin and sample conditioning, both completions retain positive mass. The width-two correct-label probability at $(1, 1)$ is strictly between zero and one, proving both assertions. □

This example does not show that the width-two correct-label probability crosses below $2/3$, and gives no uniform lower bound on its wrong-label mass or strictness after a cold limit. A specialized comparison for confidence-failure risk is not ruled out. At one hidden layer and fixed t, the raw width-kN output is an average of k independent raw width-N outputs. That averaging identity does not survive as a support-preserving comparison of globally margined sign posteriors.

6 What the bounds leave open

The general upper theorem and the stronger Boolean-circuit upper bound supply actual-posterior obstructions to depth hardness at small widths; the lower theorem supplies a dimension-dependent confidence obstruction already for a constant target. They are not matching bounds. The general upper bound retains quadratic learner-width dependence; the digital-circuit bound is linear in width at the critical scale, and the symmetry lower gives only $\Omega(\log L)$ when d is polynomial in L. The width counterexample blocks one simple interpolation principle without proving a confidence lower bound. The original question of a universal polynomial sample bound versus an exponential-depth counterexample for canonical tanh remains open.